Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative deadwood class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPPsat (µmol m⁻² s⁻¹)

57% improved | 58% improved | 56% improved | 55% improved

NEPmax (µmol m⁻² s⁻¹)

68% improved | 70% improved | 74% improved | 68% improved

ETmax (mm d⁻¹)

58% improved | 64% improved | 58% improved | 57% improved

WUE (g C mm⁻¹)

53% improved | 58% improved | 53% improved | 51% improved

ΔRMSE = RMSE(with M) − RMSE(without M)

Sites (sorted by ΔRMSE within each panel)

Relative deadwood
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 7.5 %, 37 sites)
- Mid (7.5–17.5 %, 42 sites)
- High (> 17.5 %, 22 sites)